

Bulb-in-Hand Initiation: An Illusory Altered In-Head Sensation via Colored Hemispheres, Audio Panning, and Head Tapping

Kaho Kagami
Graduate School of Nagoya City University
Nagoya, Japan
yuki.popjam1115@gmail.com

Kenri Kodaka*
Nagoya City University
Nagoya, Japan
kenrikodaka@gmail.com



Figure 1: Bulb-in-Hand Initiation

Abstract

We present *Bulb-in-Hand Initiation*, a multisensory system designed to induce an illusory in-head sensation by combining audio panning, hemispheric lighting, and head tapping. Inspired by prior work on internal localization illusions, our system uses a half-mirror XRAY environment to visually superimpose two color-emitting hemispheres onto the participant's own head. During the interaction, an experimenter shakes a handheld bulb that randomly changes color and "copies" the color to either hemisphere, while simultaneously tapping the participant's head on the corresponding side. Each hemisphere is linked to a specific frequency determined by mapping the luminance of its RGB color to a sine tone, and to a stereo-panned audio channel, creating a synchronized visual-auditory-tactile experience. A user study confirmed that this congruent multisensory interaction significantly enhanced the sensation that the internal state of the head had changed ($p < .03$), demonstrating the effectiveness of the system in evoking externally driven in-head illusions.

CCS Concepts

• Human-centered computing → Mixed / augmented reality.

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1 Introduction

At CHI 2025, we presented *Being Mosquitone* [Kodaka et al. 2025], a multisensory system that synchronized headphone audio with an HMD-based visual of a mosquito invading the user's headspace. By aligning the mosquito's lateralized position in the visual scene with audio panning in the headphones, we demonstrated that a convincing illusion of a presence existing inside the head could be evoked. While headphone output is typically criticized in acoustic engineering for inducing internal localization—often considered a flaw—we leveraged this very characteristic as a perceptual advantage, turning it into a key element of the illusion. *Being Mosquitone* effectively created the sensation of a mosquito entering from outside the head and burrowing into the ear canal. Such "in-head" illusions may hold promise for a range of applications, including externalizing intrusive thoughts, visualizing imagined content, or enhancing cognitive scaffolding for abstract thinking.

Building on these insights, our current system explores how to recreate this sensation of internal invasion in a more abstract and non-HMD setting, replacing the mosquito with the concept of "light." To achieve this, we employ an XRAY setup developed by our team [Imai et al. 2022][Kodaka and Kagami 2025], which uses a half-mirror to overlay the user's live head image with two hemispheres capable of emitting light independently in different colors. We designed an interaction in which the randomly changing light of a handheld bulb is perceived to be "copied" into the head, synchronized with a tactile tap on the corresponding side of the skull.

2 Implementation

This system builds on the XRAYHEAD setup, in which a half mirror overlays a glowing object behind it onto the participant's illuminated face. While the original version used an illuminated skull model, our installation replaces it with two color-emitting hemispheres, handcrafted from washi paper to resemble the cerebral hemispheres. Inside each hemisphere is a Philips Hue bulb connected to a local wireless network. The bulbs synchronize with the color values of a designated computer monitor, allowing each hemisphere to change color independently and dynamically.

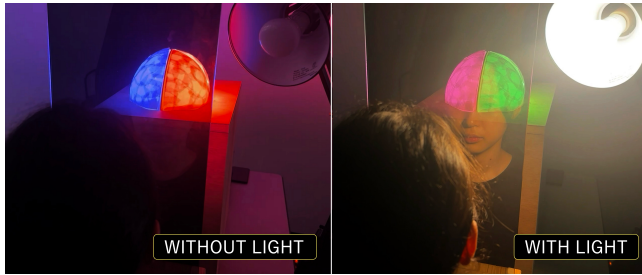


Figure 2: Visual superimposition of the participant's face and illuminated hemispheres in the XRAY environment using a half-mirror and frontal lighting.

In this system, the experimenter wears both headphones and a neck speaker. Holding a Philips Hue LED bulb in one hand, they use the other hand to tap the participant's head. The core interaction is based on a simple mapping: each time the bulb is shaken, its color changes randomly, and that color is then “copied” to one of the two artificial hemispheres displayed in the setup. Specifically, the experimenter brings the bulb close to the inner surface of one of the hemispheres (positioned behind a half-mirror), illuminating a particular area. Simultaneously, the experimenter taps the participant's head at the symmetrical location on the opposite side of the mirror. Pressure sensors are attached to the participant's index and middle fingers to detect the taps. A tap on the right side of the head triggers a color change in the right hemisphere; a tap on the left side triggers a change in the left hemisphere.

Each hemisphere is associated with a particular frequency of a sine tone. The frequency is determined by mapping the luminance of the hemisphere's RGB color (calculated using the NTSC weighting $0.299R + 0.587G + 0.114B$) to a range of 200–2000 Hz. These tones are generated in real time using Max and presented independently to the left and right channels of the headphones and neck speaker. The neck speaker is mainly for experimenter monitoring and audience feedback. For instance, when the left hemisphere is yellow and the right is blue, a high-frequency tone is played in the left ear, and a low-frequency tone in the right. If the experimenter presses two fingers across the participant's midline, the bulb's current color is simultaneously copied to both hemispheres. This synchronized update enhances the impression that a specific color has permeated the entire head. In addition to this single-color mapping, a separate rapid flip mode copies a fast alternating sequence of colors and sounds between the hemispheres.



Figure 3: Bulb shaking triggers random color changes, followed by experimenter-driven transitions that map bulb color to hemisphere lighting and stereo-panned audio.

3 Evaluation

We conducted a laboratory study to evaluate the perceptual effects of *Bulb-in-Hand Initiation*. Nineteen university or graduate students participated in a within-subjects experiment comparing two conditions: one where the stereo-panned sound matched the side of the illuminated hemisphere (congruent), and another where the sides were mismatched (incongruent). Participants rated their experience on four questions using a 7-point scale from –3 (very strongly disagree) to +3 (very strongly agree): the sense of seeing inside one's own head (Q1), internal sound localization (Q2), a feeling that the state inside the head changed due to tapping (Q3), and the sensation of light entering the head (Q4).

Average ratings were higher in the congruent condition across all questions: Q1 (0.47 → 0.53), Q2 (0.26 → 0.74), Q3 (0.37 → 1.42), and Q4 (0.11 → 0.68). A paired t-test showed a significant increase only for Q3 ($p < .03$), suggesting that congruent audio-visual-tactile alignment can enhance the illusion that something inside the head has been changed by an external action.

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